KUNE PAVITHRA
COMET FWC 69

International Institute of Information Technology Bangalore

Mathematics Olympiad Task

Q14

In fig. 6, find the area of the shaded region enclosed between two concentric circles of radii 7 cm and 14 cm where $\angle AOC = 40°$.

Use $\pi = \frac{22}{7}$.

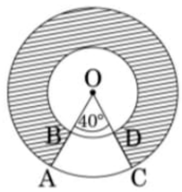

Figure 6

Q15

If the ratio of the sum of first n terms of two A.P's is

$$(7n + 1) : (4n + 27)$$

find the ratio of their mth terms.

Q16

Solve for x:

$$\frac{1}{(x-1)(x-2)} + \frac{1}{(x-2)(x-3)} = \frac{2}{3}$$

where $x \neq 1, 2, 3$.

Q17

A conical vessel with base radius 5 cm and height 24 cm is full of water. This water is emptied into a cylindrical vessel of base radius 10 cm.

Find the height to which the water rises.

Q18

A sphere of diameter 12 cm is dropped into a cylindrical vessel partly filled with water. The water level rises by 32 cm.

Find the diameter of the cylindrical vessel.

Q19

A man standing on the deck of a ship 10 m above water level observes the angle of elevation of the top of a hill as 60° and angle of depression of the base of hill as 30°.

Find:

- Distance of hill from ship
- Height of hill

Q20

Three different coins are tossed together.

Find the probability of getting:

- exactly two heads
- at least two heads
- at least two tails

Q21

Due to heavy floods in a state, thousands were rendered homeless. 50 schools collectively offered place and canvas for 1500 tents.

Each tent has:

- cylindrical base of radius 2.8 m and height 3.5 m
- conical top of radius 2.8 m and height 2.1 m

If canvas costs Rs.120 per sq.m, find the amount shared by each school.

What value is generated by this problem?

Q22

Prove that the lengths of tangents drawn from an external point to a circle are equal.

Q23

Draw a circle of radius 4 cm. Draw two tangents to the circle inclined at an angle of 60° to each other.

Q24

In Fig. 7, two equal circles with centres O and O' touch each other at X.
OO' produced meets the circle with centre O' at A.
AC is tangent to the circle with centre O at C.
$O'D \perp AC$.
Find:

$$\frac{CO}{DO'}$$

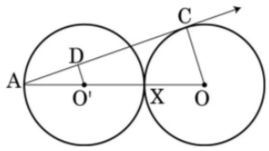

figure 7

Q25

Solve for x:

$$\frac{1}{x+1} + \frac{1}{x+2} + \frac{1}{x+4} = \frac{2}{x}$$

where $x \neq -1, -2, -4$.

Q26

The angle of elevation of the top Q of a vertical tower PQ from a point X on the ground is $60°$.
From a point Y, 40 m vertically above X, the angle of elevation of the top Q of tower is $45°$.
Find:

- height of tower PQ
- distance PX

Use $\sqrt{3} = 1.73$.

Prove the following:

1.

$$3\sin^{-1} x = \sin^{-1}(3x - 4x^3), \quad x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

2.

$$3\cos^{-1} x = \cos^{-1}(4x^3 - 3x), \quad x \in \left[\frac{1}{2}, 1\right]$$

3.

$$\tan^{-1}\frac{2}{11} + \tan^{-1}\frac{7}{24} = \tan^{-1}\frac{1}{2}$$

4.

$$2\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{7} = \tan^{-1}\frac{31}{17}$$

Write the following functions in the simplest form:

5.

$$\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right), \quad x \neq 0$$

6.

$$\tan^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right), \quad |x| > 1$$

7.

$$\tan^{-1}\left(\sqrt{\frac{1-\cos x}{1+\cos x}}\right), \quad 0 < x < \pi$$

8.

$$\tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right), \quad -\frac{\pi}{4} < x < \frac{3\pi}{4}$$

9.

$$\tan^{-1}\left(\frac{x}{\sqrt{a^2-x^2}}\right), \quad |x| < a$$

10.

$$\tan^{-1}\left(\frac{3a^2x - x^3}{a^3 - 3ax^2}\right)$$

$$a > 0, \quad -\frac{a}{\sqrt{3}} < x < \frac{a}{\sqrt{3}}$$

Find the values of each of the following:

11.

$$\tan^{-1}\left[2\cos\left(2\sin^{-1}\frac{1}{2}\right)\right]$$

12.

$$\sin\left(\sin^{-1}\frac{1}{5}+\cos^{-1}x\right)=1$$

Find x.